Series A6BAB/C

Set No. 1

प्रश्न-पत्र कोड
Q.P. Code **65/6/1**

अनुक्रमांक
Roll No.

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें ।
Candidates must write the Q.P. Code on the title page of the answer-book.

- *कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ **7** हैं ।*
- *प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें ।*
- *कृपया जाँच कर लें कि इस प्रश्न-पत्र में **14** प्रश्न हैं ।*
- ***कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें ।***
- *इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है । प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा । 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे ।*
- *Please check that this question paper contains **7** printed pages.*
- *Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.*
- *Please check that this question paper contains **14** questions.*
- ***Please write down the serial number of the question in the answer-book before attempting it.***
- *15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.*

गणित
MATHEMATICS

निर्धारित समय : 2 घण्टे — *अधिकतम अंक : 40*

Time allowed : 2 hours — *Maximum Marks : 40*

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) *This question paper contains* ***three*** *sections –* ***Section A, B*** *and* ***C****.*

(ii) *Each section is* ***compulsory****.*

(iii) ***Section A*** *has* ***6*** *short answer type I questions of* ***2*** *marks each.*

(iv) ***Section B*** *has* ***4*** *short answer type II questions of* ***3*** *marks each.*

(v) ***Section C*** *has* ***4*** *long answer type questions of* ***4*** *marks each.*

(vi) *There is an internal choice in some questions.*

(vii) *Question no.* ***14*** *is a case-study based question with 2 sub-parts of* ***2*** *marks each.*

SECTION A

Question numbers ***1*** *to* ***6*** *carry* ***2*** *marks each.*

1. Evaluate : *2*

$$\int_0^{\pi/2} \frac{1}{1+\cot^{5/2} x}\,dx$$

2. If $\vec{a} = \hat{i} + \hat{j} - 2\hat{k}$, $\vec{b} = -\hat{i} + 2\hat{j} + 2\hat{k}$ and $\vec{c} = -\hat{i} + 2\hat{j} - \hat{k}$ are three vectors, then find a vector perpendicular to both the vectors $(\vec{a} + \vec{b})$ and $(\vec{b} - \vec{c})$. *2*

3. A bag contains cards numbered 1 to 25. Two cards are drawn at random, one after the other, without replacement. Find the probability that the number on each card is a multiple of 7. *2*

4. One bag contains 4 white and 5 black balls. Another bag contains 6 white and 7 black balls. A ball, drawn at random, is transferred from the first bag to the second bag and then a ball is drawn at random from the second bag. Find the probability that the ball drawn is white. *2*

5. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are unit vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, then find the value of $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$. *2*

6. (a) Find the general solution of the differential equation

$$x \cos y \, dy = (x \log x + 1)\, e^x \, dx.$$ *2*

OR

(b) Find the value of $(2a - 3b)$, if a and b represent respectively the order and the degree of the differential equation

$$x\left[y\left(\frac{d^2y}{dx^2}\right)^3 + x\left(\frac{dy}{dx}\right)^2 - \frac{y}{x}\frac{dy}{dx}\right] = 0.$$ *2*

SECTION B

*Question numbers **7 to 10** carry **3** marks each.*

7. (a) Find the area of the region $\{(x, y) : x^2 + y^2 \leq 9,\ x + y \geq 3\}$, using integration. *3*

OR

(b) Using integration, find the area of the region bounded by the parabola $y^2 = 4x$, the lines $x = 0$ and $x = 3$ and the x-axis. *3*

8. Find : *3*

$$\int \frac{\sin x}{\sin(x - 2a)}\, dx$$

9. Find the equation of the plane passing through three points whose position vectors are $-\hat{j}$, $3\hat{i} + 3\hat{j}$ and $\hat{i} + \hat{j} + \hat{k}$. *3*

10. (a) Find the distance between the following parallel lines : *3*

$$\vec{r} = (2\hat{i} + \hat{j} - \hat{k}) + \lambda(\hat{i} + \hat{j} - \hat{k})$$

$$\vec{r} = (\hat{i} - 2\hat{j} + \hat{k}) + \mu(\hat{i} + \hat{j} - \hat{k})$$

OR

(b) Find the coordinates of the point where the line through the points $(-1, 1, -8)$ and $(5, -2, 10)$ crosses the ZX-plane. *3*

SECTION C

*Question numbers **11** to **14** carry **4** marks each.*

11. Find the equation of the plane passing through the intersection of the planes $\vec{r} \cdot (2\hat{i} + 2\hat{j} - 3\hat{k}) = 7$ and $\vec{r} \cdot (2\hat{i} + 5\hat{j} + 3\hat{k}) = 9$ and through the point (2, 1, 3). *4*

12. (a) Find : *4*

$$\int \cos x \cdot \tan^{-1}(\sin x)\, dx$$

OR

(b) Find : *4*

$$\int \frac{e^x}{(e^x + 1)(e^x + 3)}\, dx$$

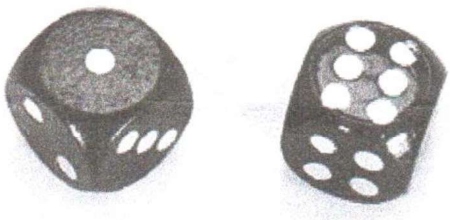
13. Find the particular solution of the differential equation

$$x\frac{dy}{dx} + 2y = x^2 \log x, \text{ given } y(1) = 1.$$ *4*

Case-Study Based Question

14. A biased die is tossed and respective probabilities for various faces to turn up are the following :

Face	1	2	3	4	5	6
Probability	0·1	0·24	0·19	0·18	0·15	K



Based on the above information, answer the following questions :

(a) What is the value of K ? *2*

(b) If a face showing an even number has turned up, then what is the probability that it is the face with 2 or 4 ? *2*